

A Partial Reduction for the Hereditary Failure of UALS

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Status

This packet records a likely valid partial result for Problem 2 in Argyros–Georgiou–Lagos–Motakis [1]. It does not solve the problem. It proves that, over complex scalars, any counterexample must lie on the unconditionally saturated side of Gowers’ dichotomy.

Source Problem

The source is arXiv:1712.07638, Section 5.5, Problem 2. The problem appears on page 36 of the source PDF.

select $n_1 \leq \dots \leq n_\ell$ such that $\|x_i \setminus \bigcup_{i=1}^{\ell} x_{n_i}\| \geq \varepsilon$ for all $i = 1, \dots, \ell$, and this yields a contradiction.

Remark 5.44. There exist Banach spaces which satisfy the UALS while this is not true for all of their subspaces. As already shown, every $L_p[0, 1]$ for $1 < p < \infty$ and $p \neq 2$, fails the UALS whereas item (ii) of Corollary 5.24 yields that it embeds in a space satisfying the property.

Another open problem in a similar context as the above remark is the following. Notice that all spaces in the previous subsection failing the UALS contain a subspace which satisfies the property.

Problem 2. Does there exist a Banach space that none of its subspaces satisfy the UALS property?

In words, the paper asks:

Does there exist a Banach space that none of its subspaces satisfy the UALS property?

Throughout this packet, “subspace” means infinite-dimensional closed subspace.

Definitions and Inputs

A Banach space X satisfies the Uniform Approximation on Large Subspaces property, abbreviated UALS, if there is $C > 0$ such that whenever $A \in \mathcal{L}(X)$, $W \subset \mathcal{L}(X)$ is convex and compact, and

every $x \in B_X$ admits $B \in W$ with

$$\|Ax - Bx\| \leq \varepsilon,$$

there are a finite-codimensional subspace $Y \subset X$ and $B \in W$ such that

$$\|(A - B)|_Y\| \leq C\varepsilon.$$

We use three standard inputs.

First, Gowers' dichotomy says that every infinite-dimensional Banach space has a subspace with an unconditional basic sequence or a hereditarily indecomposable subspace. We cite the modern arXiv exposition [3] for this statement.

Second, Ferenczi proved that if H is a complex hereditarily indecomposable Banach space, then every operator from a subspace $Y \subset H$ into H is of the form

$$T = \lambda I_{Y,H} + S,$$

where $I_{Y,H}$ is the inclusion and $S : Y \rightarrow H$ is strictly singular [2]. In particular, every operator on every subspace $Y \subset H$ is $\lambda I_Y + S$ with $S \in \mathcal{SS}(Y)$.

Third, the source paper proves that if every operator on a Banach space X is a scalar multiple of the identity plus a strictly singular operator, then X is UALS-saturated [1, Section 5.1, p. 24]. The proof is written there over real scalars, but the same argument is unchanged over complex scalars after replacing absolute values of real scalars by moduli.

Proof Intuition

The open problem asks for a space whose every subspace fails UALS. Gowers' dichotomy splits any complex Banach space into two possible hereditary directions: either one finds unconditional basic sequences, or one finds a hereditarily indecomposable subspace. The second direction cannot produce an example for this problem. Complex HI spaces have very few operators: modulo strictly singular maps, every operator is scalar. The source paper already shows that precisely this scalar-plus-strictly-singular structure is strong enough to force UALS, even in saturated form. Thus any future example must come from the unconditional side.

Partial Theorem

Theorem 1. *Let X be an infinite-dimensional complex Banach space. If X contains no unconditional basic sequence, then X contains an infinite-dimensional subspace satisfying the UALS property.*

Proof. By Gowers' dichotomy, X contains either a subspace with an unconditional basic sequence or a hereditarily indecomposable subspace. The first alternative is excluded by hypothesis. Hence X contains a complex hereditarily indecomposable subspace H .

By Ferenczi's theorem, every operator $T \in \mathcal{L}(H)$, and more generally every operator on every subspace of H , is of the form

$$T = \lambda I + S,$$

where $\lambda \in \mathbb{C}$ and S is strictly singular. Applying the scalar-plus-strictly-singular theorem from the source paper, with the identical complex-scalar proof, yields that H is UALS-saturated. In particular, H satisfies UALS. $\square$

Corollary 1. *If a complex Banach space X has the property that none of its infinite-dimensional subspaces satisfy UALS, then X is unconditionally saturated.*

Proof. Let $Z \subset X$ be an infinite-dimensional subspace. If Z contained no unconditional basic sequence, the theorem applied to Z would give a further subspace $H \subset Z$ satisfying UALS, contradicting the assumed hereditary failure of UALS. Therefore every infinite-dimensional subspace $Z \subset X$ contains an unconditional basic sequence. $\square$

Verification Notes

The proof is a direct combination of three cited theorems. No numerical or computer verification is involved.

The only scalar-field caveat is explicit: the reduction is stated for complex Banach spaces. The real-scalar analogue would require either a real HI operator-structure input strong enough to imply scalar-plus-strictly-singular, or a separate extension of the source paper's UALS-saturation argument to the possible real quotient algebras of HI spaces.

Novelty and Search Bounds

Bounded web searches were run on June 14, 2026 for the exact source question and close phrases, including:

"Does there exist a Banach space that none of its subspaces satisfy the UALS property",
 "Uniform Approximation on Large Subspaces" UALS,
 "UALS property" Banach,
 "Uniform Approximation on Large Subspaces" "complete separation".

These searches surfaced the source paper but no later paper explicitly answering Problem 2. This packet should therefore be treated as an agent-identified partial implication from known theorems, not as a literature-already-answered status packet.

Human Review Recommendation

Check the scalar-field restriction and the invocation of Ferenczi's theorem for operators on subspaces of complex HI spaces. If the run is treating all Banach spaces as real unless otherwise stated, this packet should remain a complex partial reduction rather than a real-scalar theorem.

References

- [1] S. A. Argyros, A. Georgiou, A.-R. Lagos, and P. Motakis, *Joint spreading models and uniform approximation of bounded operators*, arXiv:1712.07638.
- [2] V. Ferenczi, *Operators on subspaces of hereditarily indecomposable Banach spaces*, arXiv:math/9502207.
- [3] R. Frankiewicz and S. Kusinski, *A simple, combinatorial proof of Gowers Dichotomy for Banach Spaces*, arXiv:2106.10728.